Asem Ameen Qaid

Electrical Engineering Student | Electronics and Embedded Systems | Semiconductor and Power/Energy Interests

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Yogyakarta, Indonesia · Yemeni nationality

PROFILE

Third-year Electrical Engineering undergraduate at Universitas Islam Indonesia (CGPA 3.75/4.00) with hands-on project work in microprocessor interfacing (Intel 8086, 8255 PPI, ADC0804), digital logic design in Quartus Prime, and microcontroller systems built and tested on ESP32 and Arduino. Project experience covers sensor interfacing, calibration and measurement analysis, including benchmarking a motion sensor against 15 reference trials and bench-testing a PCB-based electrical protection system. Developing interest in semiconductor devices, microelectronics and digital hardware, alongside power electronics, power systems and renewable energy. Seeking an internship, research or technical development opportunity to strengthen hardware and electronics skills while contributing to real engineering projects.

EDUCATION

B.Sc. Electrical Engineering, Universitas Islam Indonesia, Yogyakarta

Expected 2028

- CGPA 3.75 / 4.00
- Future Global Leaders Scholarship (FGLS), 2024 – present
- Relevant laboratory coursework: Computer Systems and Microprocessors, Digital Logic Circuits (FPGA), Embedded Systems, Measurement and Instrumentation

ENGINEERING PROJECTS

Digital Hardware and Microprocessor Systems

Dryer Control System Using the Intel 8086 Microprocessor

Jan 2026

Computer System and Microprocessor · final project

- Designed a microprocessor-based controller in 8086 assembly that regulates temperature (LM35) and raises an over-pressure alarm (MPX4115A), with analog inputs digitised by an ADC0804.
- Configured an 8255 PPI in Mode 0 (control word 8AH) for display output, ADC input, push-button input and relay/buzzer control; used bit-masked output writes so inputs and outputs could share Port C without conflict.
- Verified the design in Proteus across 15 simulated test cases at setpoints of 30 °C and 100 kPa; heater and alarm behaved as specified in all cases.

Tools: Intel 8086 assembly, 8255 PPI, ADC0804, LM35, MPX4115A, Proteus

Digital Logic Circuit and FPGA Design

Laboratory coursework

- Designed and simulated digital logic circuits using FPGA development tools (Quartus Prime).

Embedded Systems, Instrumentation and Hardware Testing

Smart Electrical Panel Protection and Emergency Alert System

2026

Embedded Systems · final project

- Built a three-state (Normal, Warning, Emergency) protection system on an Arduino Nano that monitors temperature (DHT11), smoke/gas (MQ-2) and flame, and disconnects the load through a relay.
- Assembled the final circuit on a soldered PCB after breadboard debugging; passed all six structured test scenarios, with load disconnection within one 2-second polling cycle and SMS alerts delivered through a SIM800L GSM module.
- Traced false flame alarms to relay switching noise and SIM800L brown-outs to ~2 A transmit bursts; resolved them by separating sensor and actuator power domains and adding a 1000 µF buffer capacitor.

Tools: Arduino Nano, DHT11, MQ-2, IR flame sensor, SIM800L, relay, 2N2222 transistor

Speed Estimation Using the MPU6050 Accelerometer and Gyroscope

2026

Measurement and Instrumentation · final project part 2 (pair project)

- Interfaced an MPU6050 over I2C with an Arduino Nano, using a 21 Hz on-chip low-pass filter, a complementary filter ($\alpha = 0.98$) and automatic start-up calibration of all six axes.
- Benchmarked two speed-estimation methods against 15 reference trials each: step counting achieved an RMSE of 0.161 m/s against 1.106 m/s for acceleration integration, and the gap was explained by integration drift.

MQ135 Gas Sensor for CO_2 and Ethanol Concentration Measurement

2025

Measurement and Instrumentation · final project part 1

- Calibrated an MQ135 sensor (24-hour preheat, 20 kΩ load resistor, averaged R_0) and built a Python live display; measured CO_2 at 15 campus locations, from about 376 to 2,172 ppm.

IoT-Based Flood Early Warning System

2026

Disaster Mitigation · project

- Built a breadboard prototype on an ESP32 that reads rain and water-level sensors, classifies risk into three levels with LED and buzzer alerts, and publishes readings over MQTT to a Mosquitto broker and a Node-RED dashboard.

Energy Systems and Engineering Software

Industrial Energy Efficiency Copilot

Oct 2026 – present

ANGEL Innovation Hackathon 2026 · Sustainable Energy track

- Team leader of Team PowerTech (four students with a faculty supervisor); responsible for system integration, the dashboard and the MQTT data pipeline of a multi-site industrial energy monitoring platform.
- Software prototype provides baseline-based anomaly detection, energy-intensity (IKE) analysis and forecasting; detected 3 of 3 inserted anomalies with no false alarms on a synthetic 30-day dataset.
- Acquisition unit (ESP32 with a PZEM energy meter over RS485/Modbus) designed and firmware written; hardware validation planned for the programme's December 2026 stage.

Tools: ESP32, RS485/Modbus, MQTT, Python

EE Studio: Interactive Virtual Engineering Laboratory

2026

Personal project · ee-studio-red.vercel.app

- Web platform of 37 interactive labs across 7 modules, including power systems (power flow, fault analysis, protection), electronic system design (rectifiers, regulators) and renewable energy (PV, MPPT, wind, storage).
- Every model verified numerically before implementation and covered by 485 automated backend tests; built through AI-assisted development, with project scope, model verification and review of every lab done personally.

LEADERSHIP

President, UII Global

2026 – present

International student organisation, Universitas Islam Indonesia

- Lead the executive board and cross-functional teams that run cultural, educational and networking programmes for international students across multiple faculties.
- Coordinate programmes with the university administration, academic departments and external international partners.

Team Leader, Team PowerTech

Oct 2026 – present

ANGEL Innovation Hackathon 2026

- Run weekly progress reviews, a shared task board, Git-based peer review and a risk register for a multidisciplinary team.

Marketing Team Member, UII Global

2024 – 2026

- Supported event promotion, digital communications, social media marketing and international student engagement.

TECHNICAL SKILLS

Programming and Firmware	C, C++, Python, Intel 8086 assembly, Arduino IDE
Microcontrollers and Digital Hardware	ESP32, Arduino (Nano), Intel 8086, 8255 PPI, ADC0804, digital logic design in Quartus Prime (FPGA tools), PCB assembly and soldering
Electronics, Instrumentation and Communication Interfaces	Sensor interfacing (temperature, pressure, gas, flame, motion), analog-to-digital conversion, I2C, UART, RS485/Modbus, MQTT, GSM (AT commands), relay and transistor switching, isolated power-domain design
Engineering Software and Simulation	MATLAB, Simulink, Proteus, Quartus Prime, Node-RED
Testing and Engineering Analysis	Structured test scenarios, sensor calibration, digital filtering, error analysis (RMSE), simulation-based verification, numerical verification of models

RESEARCH INTERESTS (DEVELOPING)

In order of priority: (1) semiconductor devices and microelectronics; (2) digital electronics and hardware systems; (3) embedded systems and electronics; (4) power electronics; (5) power systems and renewable energy integration.

HONOURS, MEMBERSHIPS AND LANGUAGES

- First Medal, Faris Al-Ilqaa Competition · Future Global Leaders Scholarship (FGLS), UII, 2024 – present
- Founding Member, Jogja SWUG (SolidWorks User Group) · Member, Google Developer Group (GDG) · Student Representative, Almawirdy Board
- Languages: Arabic (native), English (professional working proficiency), Indonesian (intermediate)